UV Photocatalysis Reaction Sped Up by Hydrogen Peroxide

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Abstract

The presence of toxic dyes in the environment presents a significant challenge in developing an effective method to convert the dyes into less harmful substances. A vast body of literature exits on the photocatalytic degradation of methylene blue in aqueous environments in the presence of titanium (IV) oxide catalyst. Hydrogen peroxide is known to have an effect on the rate of decomposition of methylene blue.[1] In the experiment, we characterize the rate law of the photocatalytic degradation of methylene blue in solution with variable amounts of hydrogen peroxide over a period of 45 minutes as a pseudo-first order model. The absorption spectrum of the degradant in aqueous solution was recorded by a spectrophotometer, and concentration data was derived using the Beer-Lambert law. The effect of adding hydrogen peroxide in great excess and volume-dependence of the reaction rate are discussed.

1 Introduction

Methylene blue (MB) is a synthetic dye commonly used to color silk, wood, cotton, and paper, and leathers.[2] After post-dyeing and finishing, much of the MB in the textile production process ends up in the environment.[3] The presence of organic dyes such as MB in the environment as a pollutant may present significant public health and environmental risks due to its acute oral toxicity. MB is toxic, carcinogenic, non-biodegradable, and causes nausea, diarrhea, vomiting, cyanosis, shock, gastritis, jaundice, and skin/eye irritations in humans and animals; MB also presents a hazard to aquatic environments as it affects photosynthetic ability of aquatic life due to its capacity to absorb light.[2] Due to these hazards, it is important to develop more efficient and cost-effective methods of removing MB from wastewater and maintaining industrial equipment that transports fluids that contain dyes.

There are challenges developing plausible methods of treating dye-contaminated water. Bioremediation, for instance, has a limited use for this purpose due to usually being restricted to batch systems and having difficulties in upscaling and maintenance. However, the degradation of MB in packed bed reactors using biochar as packing material has been proven useful for treating wastewater at concentrations near 500 ppm[4] Another recent advancement in methods of removing MB is by adsorption using hydrogels. This method is however limited by the effect of temperature on the critical value of saturation as well as the difficulty of reusing hydrogel due to the need to desorb it.[5] In industrial applications, it is beneficial for the medium of dye removal to be reusable and cheap as well as easy to maintain.

Photocatalysis is one viable method degrading pollutants. The first photocatalytic reactors that were patented used conventional lamps in order to overcome the unreliability of natural light.[6] Since 1992, Since then, many innovations in the design of these reactors have been made to improve the exposure of reactant to the catalyst and the ease of catalyst recovery. In particular, the first reactors to use UV-LEDs as light sources were developed for the purpose of degrading dyes such as malachite green, methylene blue, and rhodameine B into less harmful compounds. The use of UV-LEDs minimize energy consumption and the generation of harmful substances due to heat.[6]

2 Background

Photocatalysis is the most reported method for the degradation of dyes.[6] Photocatalysis is a surface-dependent phenomenon in which electromagnetic radiation activates the formation of electron/hole pairs which interact with the species present in the reaction medium to carry out the degradation process.[7] The generated holes react with water to form hydroxyl radicals while the electrons interact with atmospheric oxygen to form superoxide radicals. These radicals break the $N-CH_3$ bond of the MB and allow further reactions with the broken constituents of the dye molecules to take place. As a detriment to the goal of facilitating the degradation process, electron/hole pairs can recombine, deactivating the photocatalyst. Recombination can be avoided by introducing oxidizing agents to later reaction pH conditions.[7]

Catalyst choice inside the reactor will also alter the outcome of the photodegradation. TiO_2 was used under visible light compared to Co-doped TiO_2.[8] This resulted in the $Co-TiO_2$ catalyst having 42-fold higher degradation in 90 minutes partially due to the separation of charge carrier and higher absorption rates for visible light, from the decrease in bandgap.[8]

The use of the catalyst, TiO_2 pellets, is seen to be crucial for the kinetics of the experiment as degradation of MB has been compared under UV light with the cases including the pellets having almost double the degradation over 60 minutes.[1] The MB solution spirals upwards through the UV Photocatalytic reactor containing the TiO_2 pellets to absorb the dye while in contact with UV light.

Here, we show the results of utilizing Photocatalytic reactors with TiO_2 pellets under ultraviolet light (UV) while varying the concentration of H_2O_2 to remove MB dye from water. There is usage of TiO_2 pellets in all iterations of concentration as it efficiently increases the degradation rate.[1]

3 Theory

Equations. The Beer-Lambert law,

$$A = \epsilon bC \tag{1}$$

where **A** is absorbance, ε is molar absorptivity, **b** is length of light path, **C** is concentration of observed solution, gives us the calibration curve as absorbance and concentration vary for different test runs using the UV spectrometer while molar absorptivity and length remain constant.

The material balance on methylene blue as reactant a and assuming the rate law is kC_a,

$$\frac{dC_a}{dt} = -kC_a \tag{2}$$

where $\mathbf{C_a}$ is the concentration of methylene blue, **k** is the rate constant, and **t** is irradiation time. After using the boundary condition that the initial concentration is C_{a_0} the concentration as a function of k and t can be related by the following first order rate law:

$$\frac{C_a}{C_{a_0}} = e^{-kt} \tag{3}$$

Chemical Reactions. The TiO_2 pellets inside the reactor that absorb the dye by absorbing a photon. This promotes an electron (e_{cb}^-) from valence to conduction band, and leaves behind a hole (h_{vb}^+), where if separation is maintained, by altering pH level, may result in oxidation-reduction reactions Eq. (7) and Eq. (8).[1]

$$\mathrm{TiO_2 + hv \longrightarrow e_{cb}{}^- + h_{vb}{}^+} \tag{4}$$

This is also summarized in direct absorption of the light,

$$\mathrm{Dye_{ads}{}^* + TiO_2 \longrightarrow Dye_{ads}{}^+ + TiO_2{}^-} \tag{5}$$

4 Methods

Powdered methylene blue, 5% H_2O_2, and titanium (IV) oxide in pellet form were used in the experiment. A photocatalytic reactor allowed the TiO_2 pellets to be filled in a spiral through which the MB solution was pumped, maximizing the surface area to which the solution was exposed. The light source of the reactor was a 9W ultraviolet lamp. A calibration curve was created by creating solutions of molarities 3.125e-6 M to 2.5e-5 M as samples for the photospectrometer. For the experimental trials, 1000mL volumes of solutions of 2.5e-5 M MB were created with 0 to 4.672mL of H_2O_2 added. One trial with H_2O_2 in assumed excess as according to Saquib[1] was undertaken. An exception was the trial with no peroxide being done on a 500mL volume of MB solution. The solutions were deposited into a glass vessel through which the they were cycled with the reactor using a pump while being magnetically stirred throughout the reaction. After the reaction began, samples of the solution in the reaction vessel were taken in 5 minute intervals with additional smaller intervals at the beginning of the trial. The reaction was considered to be finished at 45 minutes. The samples at each interval were deposited into cuvettes and tested with the photospectrometer to find the absorbance spectrum. The concentration of MB remaining in the solution at each time was determined using the Beer-Lambert law Eq. (1) Raw data was tabulated from the photospectrometer input using Excel, and calculations and figures were made in MATLAB.

5 Results and Discussion

Through the preliminary spectrophotometer data collected on solutions of MB with known concentrations, Fig. 1 can be plotted to display the correlation between absorbance intensity and solution concentration. The plot displays a nearly-linear relationship between absorption and concentration, which is supported by Eq. (1).

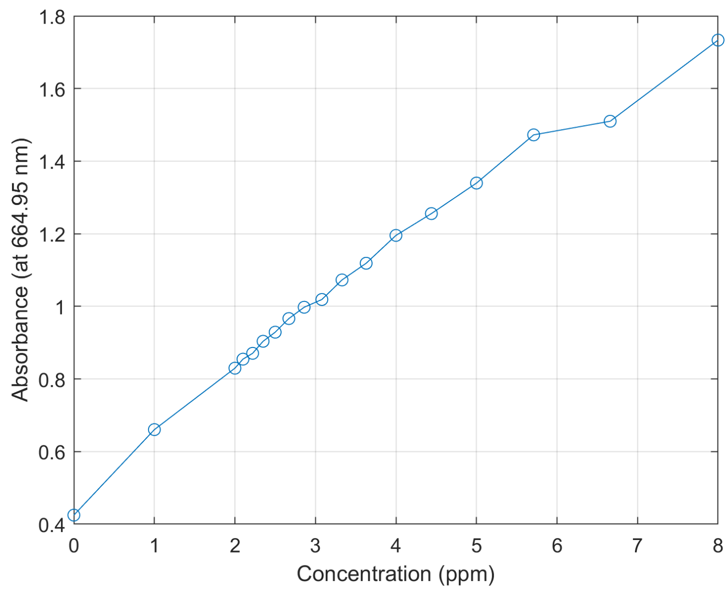

Figure 1: Absorbance as a function of known MB concentration when wavelength = 664.95 nm.

5.1 Methylene Blue Degradation in the Presence of TiO_2

Methylene blue suspended in deionized water undergoes irradiation in the presence of titanium dioxide pellets under a 9W UV-lamp. Under such conditions, the result is a decrease in solution's absorption intensity as a function of time. This normalized absorption intensity is used in the calculation of a standard calibration curve. The recorded absorption spectra (UV-absorbance vs. wavelength) aids in the selection of a suitable wavelength to best observe degradation of the dye, shown in Fig. 2. Regardless of change in time, the most significant change in absorbance, or the highest peak, occurs at approximately 664.95 nm. This wavelength falls into the range for red-orange visible light. This finding is consistent with previous research as well as color theory.

At 664.95nm, the change in concentration can be plotted as a function of time, shown in Fig. 3. The presented error range is based on manufacturer's accuracy specifications, which is $\pm 5\%$ photometric accuracy. This curve resembled that of an exponential decay, which suggests that the reaction could be modeled using Eq. (3), a first-order rate law. This equation can be used to determine the rate constant, giving more insight into the rate of decay.

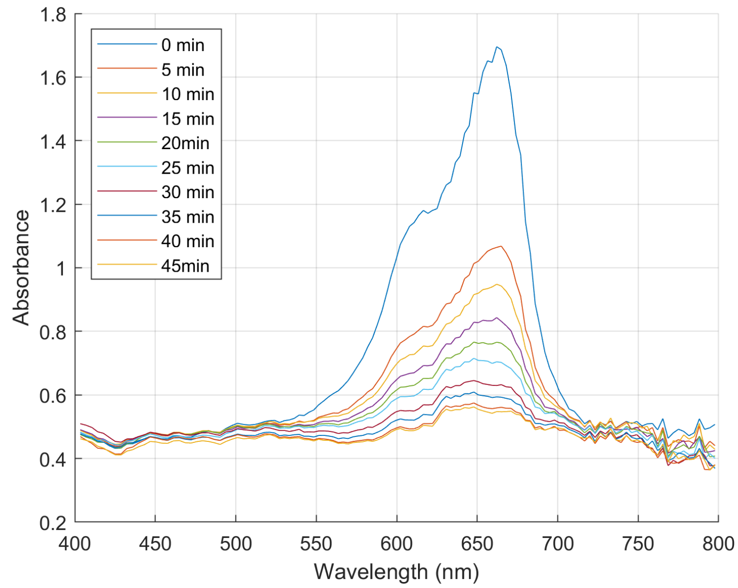

Figure 2: Change in absorbance of 2.50 $* 10^{-5}$ M MB solution as a function of time in the presence of TiO_2 and a 9W UV-light source on the absorption spectrum of visible light.

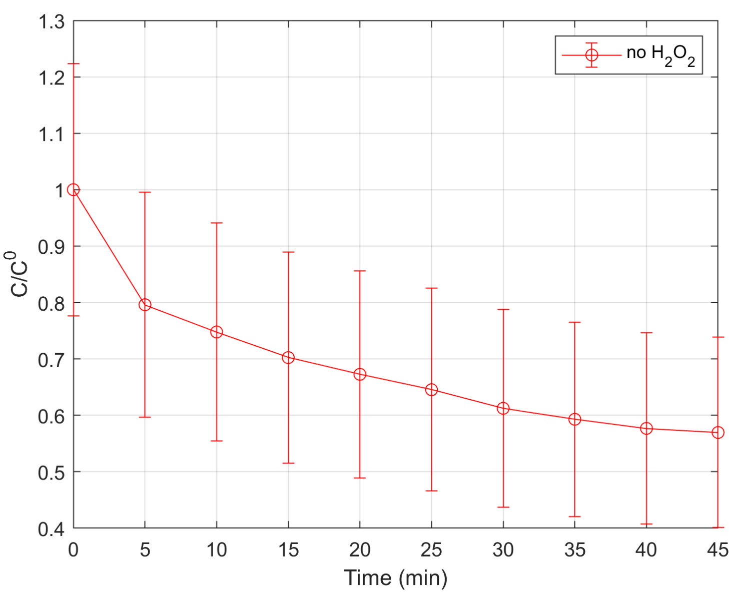

Figure 3: Change in normalized concentration of 2.50 $* 10^{-5}$ M MB solution as a function of time in the presence of TiO_2 and a 9W UV-light source, with error bars.

5.2 Effects of H_2O_2 on Methylene Blue Degradation

In the presence of hydrogen peroxide, degeneration is improved due to the formation of free radicals.[9] This enhancement is shown in Fig. 4 with error bars. When compared on the same plot in Fig. 5, there is a noticeable correlation between the increase in of H_2O_2 added and the decrease of normalized concentration as a function of time. With the trial with no H_2O_2 displaying the least amount of concentration change. Since each experiment closely resembles an exponential decay, Eq. (3) can be used to identify the rate constant for each experiment and plotted, displayed in Fig. 6. Though the rate of degeneration for MB degeneration increases with added H_2O_2, the relationship is not linear. The addition of an excessive volume of H_2O_2 does not always result in an increase in the rate constant, which can be seen in Fig. 5 in the last trial with 68 mL H_2O_2. For this experiment, the trial using 4.672 ml of H_2O_2 resulted in the largest changes in concentration over a 45-minute time period and therefore was the most optimal.

5.3 The effect of Volume on Rate Constant (k)

When using Eq. (3), it is assumed that the volume remains constant.[10] But different volumes of the same solution will likely yield concentration results when every other condition is kept constant (i.e. pump speed, residence time, molarity, reactor size, etc.). A smaller starting volume leads to the MB in solution being pumped into the reactor more often, which gives more opportunity for contact with TiO_2 and UV light source therefore speeding the rate of degeneration. The volume of the experiment with no added H_2O_2 was 500 mL, whereas all other experiments had 1000 mL. This resulted in the a very steep decay in normalized concentration and a large rate constant (k). This can be remedied by working backwards and entails adjusting k to reflect what would have been expected for the other volume, as well as Eq. (3) and the following equation:

$$-\left[\frac{1}{V}\right] * \left[\frac{dN_A}{dt}\right] = -\frac{dC_a}{dt} \tag{6}$$

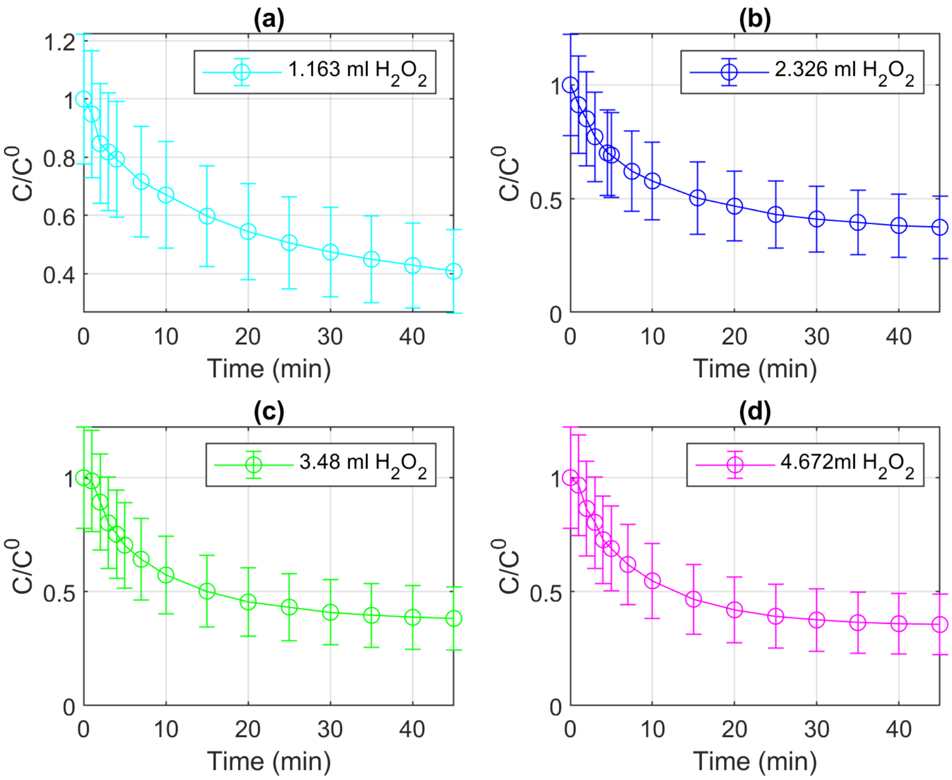

Figure 4: Change in normalized concentration of $2.50 * 10^{-5}$ M MB solution as a function of time with the addition of varying amounts of H_2O_2 with error bars.

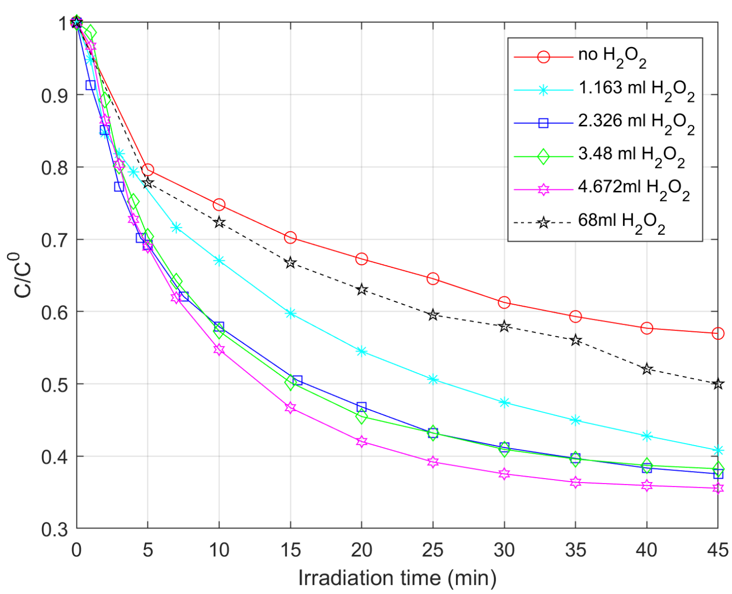

Figure 5: Change in normalized concentration of $2.50 * 10^{-5}$ M MB solution as a function of time with varying amounts of H_2O_2.

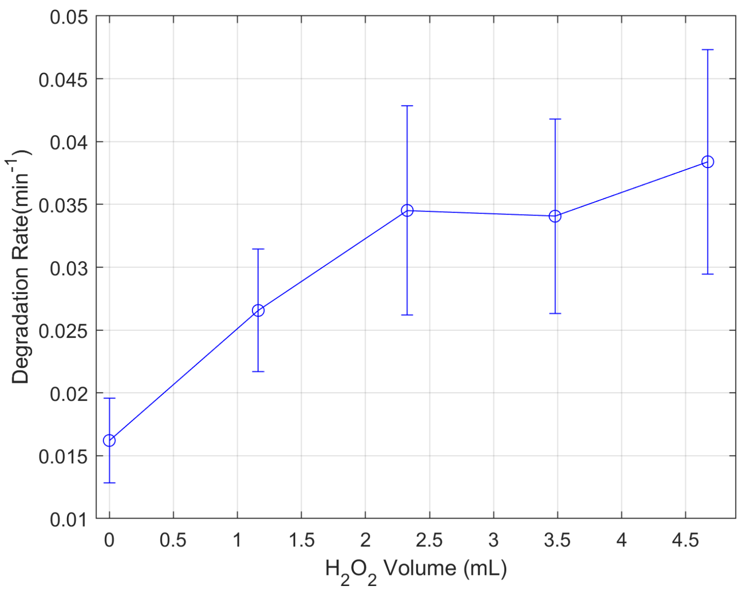

Figure 6: Degeneration rate of MB with the addition of H_2O_2.

where V is volume, $\frac{dn_A}{dt}$ is the change in moles with respect to time, and $\frac{dC_a}{dt}$ is change in concentration with respect to time. This method allows the scaling of different starting volumes to be compared side by side.

6 Conclusions

In our experiment, it was found that lowering the pH of solution using hydrogen peroxide generally accelerates the rate of reaction. Through information obtained through an anomalously prepared trial, we inferred that the photocatalytic reaction is affected by the size of the volume of solution cycled through a reactor. Further investigation in this aspect of the reaction should include the varying the residence time of the solution as well as experimenting with different types of photocatalytic reactors and types of titanium (IV) oxide pellets. Rigorous observance on the exposure of degradant to catalyst in terms of surface area should also follow. The data on the reaction rate obtained here should also be subject to comparison with mechanistically derived rate laws which separate the reaction into multiple steps where the rate-determining step can be known.

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(4) Bharti, V. Biodegradation of methylene blue dye in a batch and continuous mode using biochar as packing media. *Environ. Res.* **2019**, *171*, 356–364.

(5) Chen, M. Highly efficient and rapid adsorption of methylene blue dye onto vinyl hybrid silica nano-cross-linked nanocomposite hydrogel. *Colloid Surface A.* **2021**, *613*.

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(10) Peters, M. S.; Skorpinski, E. J. Definition of variable-volume and constant-volume reaction rates. *Journal of Chemical Education* **1965**, *42*, 329.

Appendix

1.
    - Author(s): I. Khan et al.
    - Year published: 2022
    - Journal name: Water

    (a) Photodegradation used as an advanced oxidation process for methylene blue dye removal with lower processing costs.

2.
    - Author(s): P. O. Oladoye
    - Year published: 2022
    - Journal name: Results in Engineering

    (a) Assessment of fatal toxins in water from methylene blue dye wastewater.

3.
    - Author(s): V. Bharti et al.
    - Year published: 2019
    - Journal name: Environmental Research

    (a) A Casuarina seed biochar was reusable 5 times to separate methylene blue in a packed bed bioreactor wwhere microbes were immobilized.

4.
    - Author(s): M. Chen
    - Year published: 2021
    - Journal name: Colloids and Surfaces A: Physicochemical and Engineering Aspects

    (a) Hydrogel usage in nanocomposites to promote active sites on polymer chains to expose dye molecules.

5.
    - Author(s): M. I. Din
    - Year published: 2021
    - Journal name: Journal of Cleaner Production

(a) Photocatalysts can be used to cure harmful dyes in aqueous environments.

- Author(s): R. Basumatary
- Year published: 2022
- Journal name: Korean Ceramic Society

  (a) High absorption and separation of charge carrier led to enhanced photocatalytic reaction mechanism.

  (b) Co-doped TiO_2 photocatalyst can be used in large scale in visible light.

- Author(s): M. Saquib
- Year published: 2008
- Journal name: Environmental Management

  (a) Use of artificial light is significantly faster than natural sunlight to degrade methylene blue dye.

  (b) Degradation rate found to strongly influenced by reaction pH, H_2O_2, and different forms of the photocatalyst TiO_2.

6. - Author(s): C. Xu, et al.
   - Year published: 2014
   - Journal name: Industrial & Engineering Chemistry Research

     (a) Photocatalytic Degradation of Methylene Blue by Titanium Dioxide: Experimental and Modeling Study

Derivations. Oxidation Reduction reactions for superoxide and hydroxyl radical,

$$O_2 + e_{cb}{}^{-} \longrightarrow O_2{}^{-} \tag{7}$$

$$H_2O + h_{vb}{}^{+} \longrightarrow OH + H^{+} \tag{8}$$

occurring at the catalyst surface with O_2 to produce a superoxide radical anion O_2^{-}. On the other hand, H_2O will react with the electron vacancy to form hydroxyl radical OH.